Almouthana Taha Khalfallah

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EDUCATION

National Higher School of Engineers of Tunis – ENSIT — Tunis, Tunisia
Research Master's Degree in Data Science — 2026 – Present
Engineering Degree in Applied Mathematics and Modeling – **1st in Cohort** — 2024 – Present

Preparatory Institute for Engineering Studies of Tunis – IPEIT — Tunis, Tunisia
Preparatory Degree in Physics and Chemistry (PC) — 2022 – 2024

EXPERIENCE

SIGA — Tunis, Tunisia
AI Intern — Jun 2026 – Jul 2026
- Developed an HR agent at SIGA automating leave management, employee requests, and document verification workflows.
- Delivered the system end-to-end within Oracle APEX, from backend logic to OCR/LLM-based document validation.

EXCELLIA Solutions — Tunis, Tunisia
Intern — Jun 2025
- Explored RAG systems and Streamlit for AI application deployment, prototyping ML models using Scikit-Learn, Pandas, and Matplotlib.
- Observed Agile workflows (JIRA, GitLab) and completed practical labs on Azure AI services (AI-900, AI-102).

SELECTED PROJECTS

MindEase | TypeScript, React, Vite, WebExtension (MV3), Mistral AI, Q-Learning, KaTeX, OCR
- Built an AI-native browser extension restructuring web pages into multi-modal representations for neurodiverse learners.
- Designed an on-device Q-learning RL agent (81 discrete states, 9 actions) dynamically tuning content chunking and complexity.
- Integrated LLM-based annotation, automated diagram generation, and OCR-driven extraction, synthesizing context across browser tabs.

HR Management Agent | PL/SQL, Oracle APEX, Python, OCR, LLM
- Built an automated HR assistant within Oracle APEX to streamline employee leave applications and document verifications.
- Integrated an OCR and LLM pipeline with PL/SQL database routines to cross-reference document data against internal policies.

LocMemory | Python, NetworkX, PyTorch, React, FastAPI, Supabase, Ollama
- Designed a local memory system for LLMs combining semantic search with a knowledge graph for structured retrieval.
- Implemented a PPO-based reinforcement learning policy to dynamically prioritize and filter candidate memories at inference.

MindMate | React, FastAPI, Supabase, Python
- Built a full-stack productivity workspace featuring an LLM companion for sentiment analysis and intelligent task extraction.
- Designed a RESTful backend with FastAPI and Supabase for real-time synchronization, session state, and user authentication.
- Deployed the application on Vercel/Render with a CI/CD pipeline via GitHub Actions and a Supabase-backed PostgreSQL database.

Micrograd Reconstruction | Python
- Built an autograd engine implementing backpropagation via reverse-mode automatic differentiation, with a small neural network library on top.

RESEARCH & OPEN SOURCE ACCOMPLISHMENTS

MindEase Research Paper | Co-Author
- Co-authoring a research paper on the 3-layer cognitive architecture and Q-learning results behind MindEase (GCD4F 9 Global Finalist).

WnR | Open Source Contribution – github.com/RoderickQiu/wnr (1.1k⋆)
- Added Arabic/French localization with RTL support and a Unicode bidi-isolation fix; merged into a 1.1k-star Electron app.

ACTIVITIES

GCD4F 9 (Global Competition on Design For Futures) — Jan 2026 – Oct 2026
- Qualified for the Global Finals in Beijing, China.

NERDATA ENSIT CLUB — Tunis, Tunisia
- Participated in organizing DataFest 1.0, Inspiring Genius, and The Upside Hack AI-Hackathon 1.0. — Sep 2024 – Present
- Led a workshop on Git and GitHub.

Hackathons — Tunis, Tunisia
- CodeCrafters, GCD4FE 8, Seneca Hack 1.0, Geeks Hack 5.0, Hack4Ucar 1.0, GCD4F 9

GEO-OLYMPIAD — Tunis, Tozeur, Tunisia
- Ranked 4th in Tunis region and qualified for the national finals of the Tunisian Geography Olympiad. — Mar 2022

SKILLS

Programming Languages: Python, C, C++, MATLAB, SQL, PL/SQL, HTML, CSS, JS, TS, WinDev (WLanguage)
Libraries & Data: NumPy, Pandas, Scikit-Learn, Matplotlib, Seaborn, TensorFlow, PyTorch, OpenCV, HuggingFace, Ollama, NetworkX
Technologies & Tools: Oracle APEX, PostgreSQL, Azure, FastAPI, React, Supabase, Vercel, Netlify, Render, Git, GitHub, GitHub Actions, Streamlit, VS Code, Jupyter Notebook, Linux, Docker, Notion, JIRA
Concepts: Machine Learning, Deep Learning, RAG, Reinforcement Learning, Knowledge Graphs, Agentic Systems, MCP, Tokenization
Languages: Arabic (Native), French (Fluent), English (Fluent – TOEIC 900/990)